But lazy Einstein was not. He was just selective. Some parts of the coursework he absorbed thoroughly; others he ignored, preferring to spend his time on self-directed study and thinking. Thinking was fun, joyful, and satisfying; on his own he could learn about the "new" physics, the physics that Heinrich Weber omitted from all his lectures.

Newton's Absolute Space and Time, and the Aether

The "old" physics, the physics that Einstein *could* learn from Weber, was a great body of knowledge that I shall call *Newtonian*, not because Isaac Newton was responsible for all of it (he wasn't), but because its foundations were laid by Newton in the seventeenth century.

By the late nineteenth century, all the disparate phenomena of the physical Universe could be explained beautifully by a handful of simple *Newtonian physical laws.* For example, all phenomena involving gravity could be explained by *Newton's laws of motion and gravity:*

- Every object moves uniformly in a straight line unless acted on by a force.
- When a force does act, the object's velocity changes at a rate proportional to the force and inversely proportional to its mass.
- Between any two objects in the Universe there acts a gravitational force that is proportional to the product of their masses and inversely proportional to the square of their separation.

By mathematically manipulating[1] these three laws, nineteenth-century physicists could explain the orbits of the planets around the Sun, the orbits of the moons around the planets, the ebb and flow of ocean tides, and the fall of rocks; and they could even learn how to weigh the Sun and the Earth. Similarly, by manipulating a simple set of electric and magnetic laws, the physicists could explain lightning, magnets, radio waves, and the propagation, diffraction, and reflection of light.

Fame and fortune awaited those who could harness the Newtonian laws for technology. By manipulating the Newtonian laws of heat, James Watt figured out how to convert a primitive steam engine devised by others into